

Guide: Why a Fixed Number of Returns Is Insufficient

Collatz proof project

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What Lean proves

Choose any finite number of returns to the progression two modulo nine. There is a positive integer whose next that many consecutive returns all increase. They are actual consecutive visits, not a selected subsequence. Each return takes four ordinary shortcut Collatz steps.

The construction

Use the seeds defined by $n_0 = 2$ and $n_{R+1} = 144n_R + 299$. The seed n_R follows the odd, odd, even, odd pattern for R blocks. Each block returns to the progression and has coefficient $27/16 > 1$. The intervening remainders are eight, eight, and four, so there are no hidden visits between the displayed returns. The proof tracks divisibility of $11n + 23$ by a power of sixteen. Every four-step block consumes one power. The recurrence supplies as many powers as the requested finite experiment needs.

What this rules out

We cannot prove universal descent by taking one fixed power of the return map: every proposed fixed number of returns has an integer counterexample. A proof still might establish descent after a number of returns depending on the starting seed. That remains open.

The result does not construct an infinite growing orbit. A larger requested length uses a different, larger seed. Exchanging those two quantifiers would be an invalid inference.

Changing branches can reset the divisibility count

The repeated-branch proof consumes powers of two in $11n + 23$. It might suggest counting those powers as a resource. But switching from the four-step pattern 1101 to the three-step pattern 101 can reset the count arbitrarily high. Lean now certifies a family with incoming count five and outgoing count $6r$, for any positive r . Thus a proof cannot assume a fixed bound on the reset even for these two fixed branches. It must account for the additional divisibility encoded in the seed. This does not rule out all more sophisticated decreasing measures.

How this fits the earlier results

The earlier sixteen-step theorem applies when a first return contracts. Here the chosen returns expand, so the theorems do not conflict. Likewise, the equivalence between sampled contraction and nondivergence does not promise a uniform stopping bound. No proof or disproof of Collatz is claimed.

Verification

From `lean/`, run `lake build Collatz.GrowingReturns`. The theorem is included in the selected axiom audit. The accompanying paper gives the exact construction and the quantifiers that limit its scope.